MARCO ANTONIO ADAME RODRÍGUEZ

IoT & EMBEDDED SYSTEMS ENGINEER

Digital Signal Processing · Scientific Software

CONTACT

ragnarockgames86@gmail.com

+52 747 334 2910

github.com/Ryukert

linkedin.com/in/marco-antonio-adame-rodríguez

ORCID 0009-0003-2361-4945

Mexico (GMT-6)

Available for remote work with US and European overlap

PROFILE

Computer Engineer and MSc candidate in Engineering, Innovation and Technological Development (UAGro, expected November 2026). I build end-to-end IoT systems: MEMS sensor, embedded firmware, real-time acquisition, Linux backend, cloud storage and visualization dashboard. Lead author of a peer-reviewed paper in Revista Vínculos, co-author of two more, and speaker at the 15th International Computing Conference (Tunja, Colombia, 2025).

SKILLS

Languages: Python, C/C++ (embedded), JavaScript, SQL, PHP, Bash.

Data & signals: NumPy, Pandas, SciPy, Matplotlib, FFT, PSD/Welch, RMS, digital filters, coherence, phase, time series.

IoT & hardware: Raspberry Pi, ESP32, Arduino Nano RP2040 Connect, MEMS sensors, I2C, UART, SPI, WiFi, Ethernet.

Backend & cloud: FastAPI, REST APIs, HTTP/HTTPS, webhooks, Supabase, PostgreSQL, Firebase.

Systems & DevOps: Linux (Ubuntu Server), Docker, systemd, SSH, VNC, Git/GitHub, Vercel.

Visualization: PyQt5, web dashboards, scientific plotting, CSV/JSON/PDF export.

EXPERIENCE

Aug 2024 – Present ■ **UAGRO · MIIDT GRADUATE PROGRAM**

Lead Researcher and Developer — SHM-IoT System

- **Designed and implemented end to end** a low-cost structural health monitoring solution for civil works, field-validated against commercial GEA and P-Alert equipment.
- Integrated acquisition nodes on Arduino Nano RP2040 Connect and Raspberry Pi 4 with MPU9250/MPU6050, LSM6DSOX and ADXL335 MEMS sensors over I2C, SPI and UART.
- **Implemented synchronized dual-core acquisition at 200 Hz,** with FFT analysis over 2048-point windows and 0.98 Hz frequency resolution.
- Built the DSP pipeline: RMS, FFT, power spectral density (Welch), dominant frequency, coherence, phase analysis and cross-correlation.
- Developed Python and PyQt5 applications for acquisition, live monitoring and analysis, with session control and automatic crash recovery.
- Stood up a centralized multi-site receiver over HTTP/HTTPS with local buffering that prevents data loss during connectivity outages.
- Developed REST APIs in FastAPI and administered continuously running Ubuntu servers: systemd, SSH/VNC, Docker, backup and synchronization.
- Published web dashboards that let users visualize, filter and download data from both custom nodes and commercial sensors.
- Validated the system on a biaxial shaking table, university buildings, a botanical garden, a high school campus and lightweight experimental structures.

PUBLICATIONS & TALKS

- **Adame Rodríguez, M. A., et al. (2026).** Ambient Vibration Measurement System on a Biaxial Vibrating Table. Revista Vínculos, 22(1). [Lead author]
- **Alcaraz Vázquez, M., Adame Rodríguez, M. A., et al. (2025).** Feature-importance-based attribute selection for dengue case classification in Mexico. Revista Vínculos, 22(2). [Co-author]
- **García Guevara, D. S., ... Adame Rodríguez, M. A., et al. (2025).** IoT-based biological sample shaker with temperature control. Revista Vínculos, 22(2). [Co-author]
- **Conference talk:** "Ambient Vibration Measurement System on a Biaxial Vibrating Table". 15th International Computing Conference (CICOM 2025), Tunja, Colombia, October 2025.

PROJECTS

SHM-IoT MIIDT — Low-cost structural monitoring · Python, C++, Raspberry Pi, PyQt5
Full architecture for acquisition, storage, analysis and visualization of accelerations in civil structures, field-tested against commercial sensors.

Geo-referenced SHM monitoring network · JavaScript, Vercel
Web dashboard showing location and status of distributed monitoring nodes, continuously deployed and public. github.com/Ryukert/SHM_GPS_NETWORK_DASH_elite

Supabase sensor dashboard · JavaScript, Supabase, PostgreSQL
Web application for submitting, querying and visualizing measurements stored in the cloud. github.com/Ryukert/MIIDT_SUPABASE_DASH

ESP32 telemetry APIs · Python, FastAPI
REST services for receiving, persisting and querying acceleration data, plus scientific plotting tools. github.com/Ryukert/APIAceleraciones

LANGUAGES

- Spanish: native.
- English: intermediate (B1), TOEFL ITP 470 (2025). Fluent reading of technical documentation.

CERTIFICATIONS

- Data Science Essentials with Python — Cisco / UAGro (Dec 2025).
- Cybersecurity Essentials — Cisco (Mar 2022).
- NDG Linux Unhatched — Cisco (May 2022).
- CCNAv7: Introduction to Networks — Cisco (Jun 2021).

INTERESTS

- Internet of Things and embedded systems.
- Digital signal processing and scientific software.
- Structural monitoring and low-cost instrumentation.
- Backend development and data infrastructure.

EDUCATION

Aug 2024 – Nov 2026 ■ **UNIVERSIDAD AUTÓNOMA DE GUERRERO**
MSc in Engineering, Innovation and Technological Development

- Thesis: "Development and implementation of a low-cost IoT system for structural health monitoring in civil works". In final stage.

2018 – 2022 ■ **UNIVERSIDAD AUTÓNOMA DE GUERRERO**
BSc in Computer Engineering

- Degree awarded 2023. Federal professional license no. 13426767.

ADDITIONAL EXPERIENCE

Jun 2023 – Jun 2024 ■ **MINISTRY OF LABOUR (MEXICO)**
Jóvenes Construyendo el Futuro — Publishing department

- 12-month vocational training programme in digital production and printing.

Aug 2022 – Jan 2023 ■ **35TH MILITARY ZONE (MEXICAN ARMY)**
Professional internship — IT area

- 450 certified hours. Computer equipment support and maintenance in an institutional environment with strict operational discipline.

Feb 2022 – Jul 2022 ■ **IGNACIO MANUEL ALTAMIRANO SECONDARY SCHOOL**
Social service — Computing area

- 500 certified hours. Technical support and community-sector educational support.